MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5977

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, subminiature, medium Mu

Outline - - - 3-1 (EIA)

Base - - - E8-10

Envelope - - - T3

Cathode - - - Coated unipotential

Base connections:

Pin No.	- - -	1	2	3	4	5	6	7	8
Element	- - -	g	nc	h	nc	k	h	nc	p

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ib	Pp	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	ft
Maximum:	6.6	180	0	200	---	1.1	22	1.1	220	---
Minimum:	6.0	---	-55	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	0	270	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS (see note 2)			UNIT
						MIN	BOGEY	MAX	
	Quality conformance inspection, part 1								
1256	Electrode current(1) (anode)		---	---	Ib	---	10.0	---	mAdc
1256	Electrode current(1) (anode)		0.4	II	Ib	7.0	---	13.0	mAdc
1256	Electrode current(2) (anode)	Ec = -13.5 Vdc; Rk = 0	0.4	II	Ib	---	---	100	µAdc
1266	Total grid current	Rg = 1.0 Meg	0.4	II	Ic	0	---	-0.4	µAdc
1301	Heater current		---	---	If	---	150	---	mA
1301	Heater current		0.4	II	If	140	---	160	mA
1306	Transconductance(1)		---	---	Sm	---	4,500	---	µmhos
1306	Transconductance(1)		0.4	II	Sm	3,700	---	5,300	µmhos
1336	Heater-cathode leakage		0.4	II	Ihk	---		5.0	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		2.5	I	R	100	---	---	Meg
1031	Low-frequency vibration	Rp = 10,000 ohms; Ck = 1,000 µF; F = 40 Hz; 15 G	2.5	I	Fp	---	---	50	mVac
1246	Audio frequency noise	Esig = 40 mVac; Ec = -4 Vdc; Rg = 0.1 Meg; Rp = 0.01 Meg; Rk = 0(see note 1)	2.5	I	---	---	---	---	---
1266	Grid emission	Ef = 7.5 V, Ec = -13.5 Vdc; Rg = 1.0 Meg; Rk = 0(see note 4)	2.5	I	Isg	0	---	-0.5	µAdc
1306	Transconductance(2)	Ef = 5.7 V	2.5	I	ΔSm Ef	---	---	15	%
1316	Amplification factor		6.5	16	Mu	13.5	16.0	18.5	---
1331	Direct-interelectrode capacitance	No shield	6.5	Code F	Cpg Cin Cout	1.0 1.5 0.6	--- --- ---	1.6 2.5 1.0	pF pF pF
1116	Lead fatigue		2.5	Code F	---	---	---	---	---
2126	Glass envelope strain		6.5	I	---	---	---	---	---
1041	Shock	450 G, Ehk = 100 Vdc; Rg = 0.1 Meg	---	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 2	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS (see note 2) MIN	BOGEY	MAX	UNIT
	Quality conformance inspection, part 2 - continued								
---	Post-shock and vibration-fatigue end points:								
1031	Low-frequency vibration		---	---	Ep	---	---	200	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	---	20	µAdc
1306	Change in transconductance (1) of individual tubes		---	---	ΔSmt	---	---	20	%
1105	Permanence of marking		---	---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	COMBINED SAMPLES	SYMBOL	LIMITS MIN	MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life test	Ef = 7.0 V; 1 minute "on" 4 minutes "off" Ehk = 140 Vac; Ec = Eb = 0 (see note 5)	2.5	Code H	---	---	---	---	---	---
1516	Stability life	Ehk = 200 Vdc; Rg = 1.0 Meg; TA = room	1.0	I	---	---	---	---	---	---
	Stability life-test end point:									
1306	Change in transconductance (1) of individual tubes		---	---	---	---	ΔSmt	---	10	%
1501	Intermittent life (room temperature)	Stability life-test, or equivalent conditions; TA = room (see notes 6 and 7)	---	---	---	---	---	---	---	---
---	Intermittent life-test end point: (room temperature) (500 hr)									
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
---	Intermittent life-test end point: (room temperature) (1,000 hours)									
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
1501	Intermittent life	Stability life-test conditions; Tf = 220°C (min) (see note 8)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points: (500 hours)									
---	Inoperatives	(see note 3)	---	---	1	3	---	---	---	---
1301	Heater current		---	---	1	3	If	138	164	mA
1266	Grid current		---	---	1	3	Ic	0	-0.6	µAdc
1306	Change in transconductance (1) of individual tubes		---	---	1	3	ΔSmt	---	20 *	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	15	%

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					TEST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 - Continued									
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	uAdc
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
1306	Transconductance (1), average change		---	---	---	---	Avg ΔSmt	---	15	%
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points: (1,000 hours)									
---	Inoperatives	(see note 3)	---	---	1	3	---	---	---	---
1301	Heater current		---	---	1	3	If	138	164	mA
1266	Grid current		---	---	1	3	Ic1	0	-0.6	uAdc
1306	Change in transconductance (1) of individual tubes		---	---	1	3	ΔSmt	---	25	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	20	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	uAdc
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---

NOTES:

1. The rejection level shall be set at the VU meter reading obtained during calibration.

2. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-months rule shall apply. MIL-STD-105, sample code letter E, shall apply.

3. An inoperative, as referenced in life test, is defined as a tube having one or more of the following defects: discontinuity, permanent shorts, or air leaks. (See method 1201.)

4. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ec Vdc	Eh Vdc	Rk Ohms	Rg Meg
7.5	0	100	270	1.0

5. The no-load to steady-state full-load regulation of the heater-voltage supply shall be not more than 3.0 percent. This test shall be made on a lot-by-lot basis.

6. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

7. Eligibility for early release shall be as follows: Early release may be made at 500 hours providing no lot failure due to this life test or the regular 1,000-hour intermittent life test has occurred in the preceding three consecutive lots and the lot under consideration is acceptable at 500 hours.

   One lot failing the regular 1,000-hour intermittent life test or the 1,000-hour intermittent (room temperature) life test shall result in loss of eligibility for the early release procedure.

   All intermittent (room temperature) life-test samples shall be operated to 1,000 hours.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226) will be satisfied if a tube having bogey Ib (+5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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